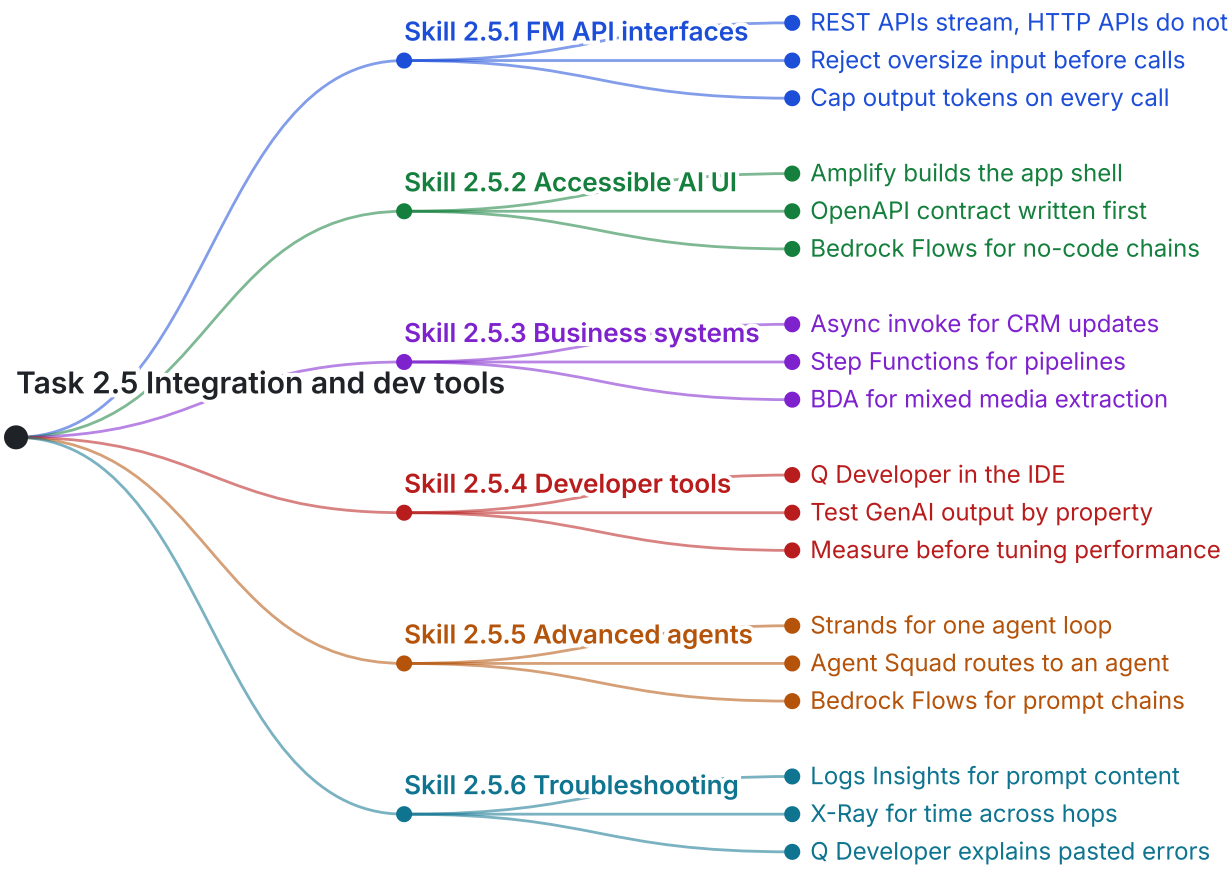
Memory sheet: Task 2.5 Implement application integration patterns and development tools

Domain 2, weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
A reply must arrive in pieces through API Gateway	REST API with STREAM transfer mode and a proxy integration	The API is an HTTP API, which cannot stream the response
The client and the server both send messages during one session	WebSocket API with routes for each message type	One-way text only, where REST streaming or server-sent events fit
The caller needs only an acceptance now and no result in the response	Asynchronous Lambda invocation with an on-failure destination	The caller needs the function result in the same response
A backlog must drain at a set rate across many messages	SQS queue with Lambda consumers and a dead-letter queue	One event needs one function call and a failure record only
Analysts must change the steps without writing orchestration code	Bedrock Flows with a published version and an alias	The run needs waits, per item retries, or audit history
A workflow must pause for a person and then resume	Standard Step Functions workflow with a callback task	The Express workflow type, which lacks the callback pattern
Many items need a summary each, with per item retries and run history	Standard workflow with Distributed Map and a Catch per item	One Lambda function looping over every item, which gives no per item status
Which agent should answer a request, with shared context	AWS Agent Squad, which classifies intent and routes the query	One agent that picks its own tools, which is the job of Strands

Etch lines

1. Lambda response streaming runs natively on Node.js managed runtimes.
2. Logs Insights shows what the prompt and response said, while X-Ray shows timing.
3. A Standard workflow with Distributed Map fits many items with per item retries.
4. Express workflows lack the callback pattern, so they cannot wait for a person.
5. Amazon Q Developer output still needs tests, review, and a security scan.
6. Measure the slow hop with traces before changing code or memory.

Flash questions

- Q: Can Lambda deliver the same asynchronous event more than once? A: Yes, so the handler must tolerate duplicate events.
- Q: How long can an Express workflow run at most? A: Five minutes.
- Q: Which Step Functions type can pause a run for a person through a task token? A: Standard, because it supports the callback pattern.
- Q: Should a browser call Bedrock directly in a sound design? A: No, the browser calls the API and the backend calls Bedrock.
- Q: Which AWS tool shows how long each hop took across API Gateway, Lambda, and a model call? A: AWS X-Ray, through its trace map and timeline.
- Q: Which service lets analysts edit a chain of prompts without code? A: Bedrock Flows, published as a version behind an alias.

- Q: Which framework routes a request to the right agent? A: AWS Agent Squad, since Strands runs the tool loop of one agent.
- Q: Does Amazon Q Developer replace code review, tests, and the security scan? A: No, it shortens the loop, and those checks still run.

Source: Task 2.5 sections for Skill 2.5.1 to 2.5.6, X versus Y decisions, Read a long question, Distractor patterns, and Drill. AWS docs and the study guide sections above, no practice-test text.